National University of Computer and Emerging Sciences

Lahore Campus

Computer Organization and Assembly Language Lab (EL2003)

Sessional-I Exam

Date:

Lab Instructor(s)

Ms. Anmol

Total Time (Minutes): **120**

Total Marks: **35**

Total Questions: **2**

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Roll No Section Student Signature

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Attempt all the questions.

1. **Create one file per question (Q1.asm, Q2.asm). Zip both with the screenshots, name the zip with your roll number (e.g. 23L-1234.zip) and submit it as told by the invigilator.**
2. **Discussion or sharing of code is strictly prohibited. Matching codes are treated as copying for all students involved. Violation of any instruction will result in negative marking.**

CLO 3: Develop assembly language programs to solve computational problems using appropriate programming techniques.

Q1 – Second Largest Element Using Nested Loops [15 Marks]

Write an 8086 Assembly Language program to find the largest and second largest elements from an array of 10 integers.

Requirements

1. Declare an array containing 10 integers.
2. Use nested loops to traverse and compare the elements of the array.
3. Find the largest and second largest values without sorting the array.
4. Store the results in two memory variables:
    - Largest
    - SecondLargest
5. Your program should work correctly for different array values. Assume that the array contains at least two distinct values.
6. You cannot sort the array.

Example

For the following array:

```
array DW 12, 45, 7, 23, 56, 18, 34, 9, 41, 27
```

The expected result is:

- Largest = 56
- SecondLargest = 45

Q2 – Parameterized Diamond Using a Subroutine [15 Marks]

Write an 8086 Assembly Language program that contains a subroutine named `DrawDiamond` to display a diamond pattern on the screen using the characters `/` and `\`.

The position and size of the diamond must be controlled by three values: `Row`, `Col`, and `X`.

- **Row:** Starting row on the screen.
- **Col:** Starting column on the screen.
- **X:** Number of rows from the top vertex to the middle row, including both rows.

Requirements

1. Declare and store `Row`, `Col`, and `X` in memory.
2. Pass all three values to `DrawDiamond` using the stack.
3. Implement `DrawDiamond` as a separate subroutine using `CALL` and `RET`.
4. Use loops to draw the diamond, with its size and position determined by the supplied parameters.
5. Changing `Row`, `Col`, or `X` should change the diamond's position or size without changing the subroutine's logic.
6. Display the diamond at the specified screen position.

Example 1

Suppose the following values are provided:

```
Row = 5
Col = 10
X = 3
```

The top of the diamond should begin at row 5, column 10.

Expected pattern:

```
  /\
 /  \
/    \
\    /
 \  /
  \/
```

The spaces shown before the pattern represent its horizontal position. The pattern has 3 rows in the upper half and 3 rows in the lower half.

Example 2

If the parameters are changed to:

```
Row = 10
Col = 20
X = 5
```

The same subroutine should draw a larger diamond beginning at row 10, column 20.



Important Instructions

- Assume the screen has 25 rows (0–24) and 80 columns (0–79).
- `Row` and `Col` specify the position of the top vertex of the diamond.
- Ensure the entire diamond remains within the screen boundaries.
- Each side of the diamond should contain exactly `X` slash characters.
- Use stack-based parameter passing; do not access the parameter values as global variables inside `DrawDiamond`.